



Society of Business

Open Project 2026 - Analytics

Agentic AI-Based Dynamic Tariff Optimization for EV Charging Networks

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Chemical Sciences



EXECUTIVE SUMMARY:

Problem:

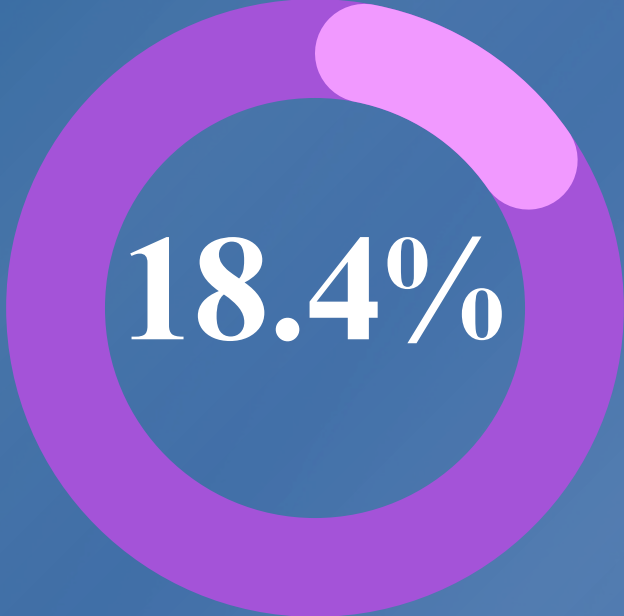
Static ₹15/kWh pricing ignores real-time demand; 62.1% charger time wasted

Solution:

3-agent AI system — predict demand, set dynamic tariff, self-correct via feedback loop

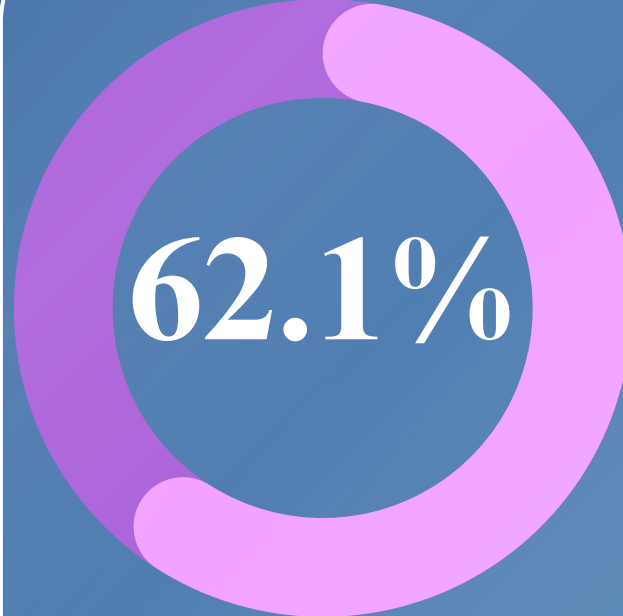
Result:

$R^2=0.955$ demand model with Stacking Ensemble — Random Forest + XGBoost + LightGBM with Ridge meta-learner



18.4%

Night revenue



62.1%

Idle time

KEY NUMBERS:

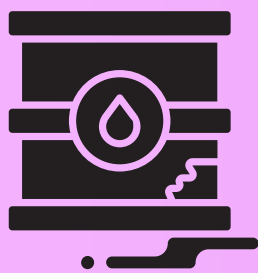
- 14,848 ACN sessions
- 247 grid zones
- 3 AI agents

PROBLEM STATEMENT & MOTIVATION:

Why static tariffs fail EV networks?



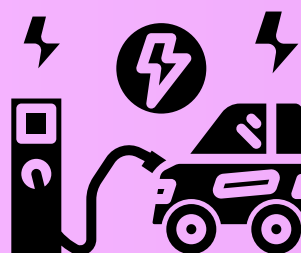
Grid overload at peak hours — flat pricing gives users zero incentive to shift demand



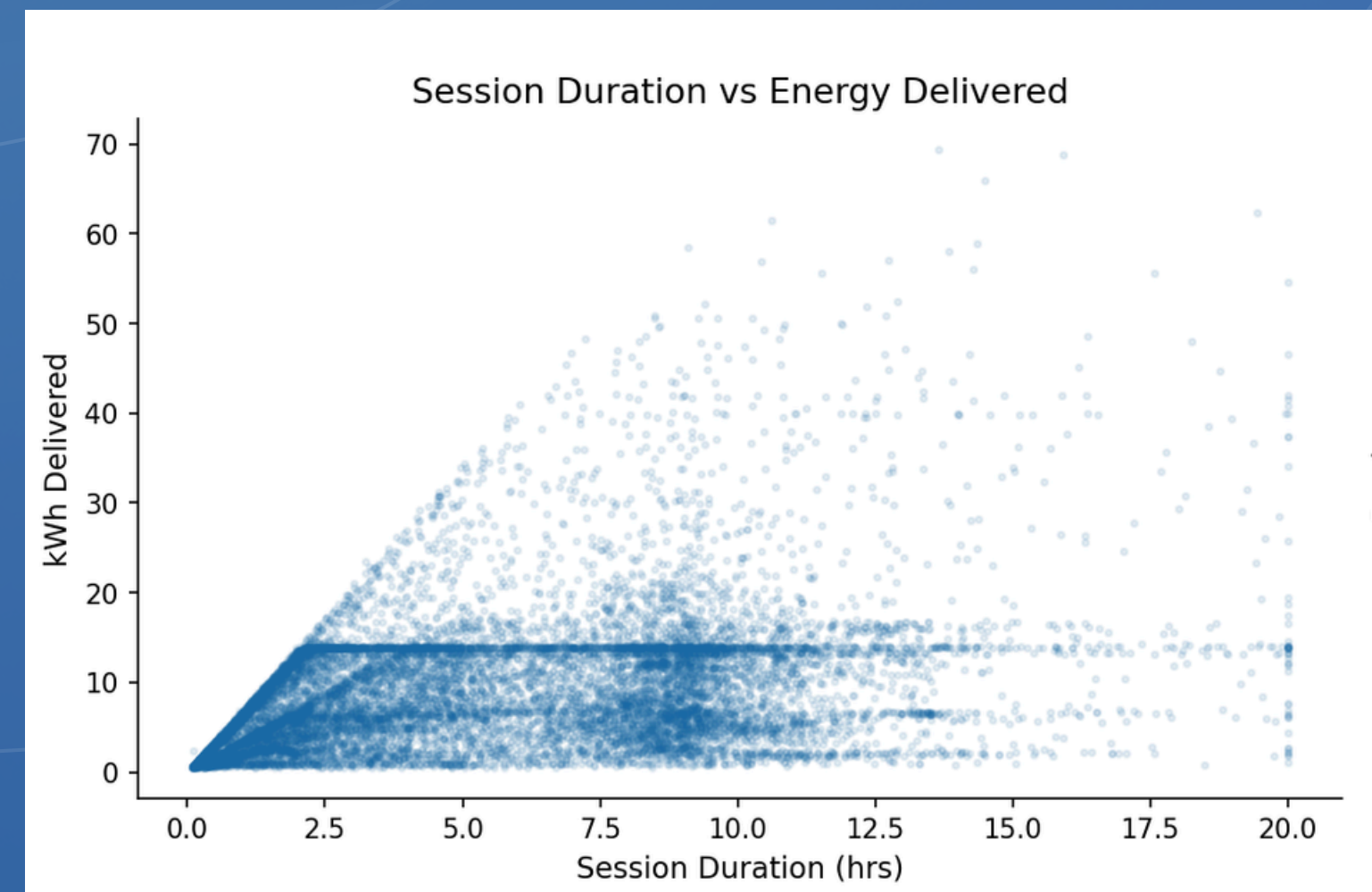
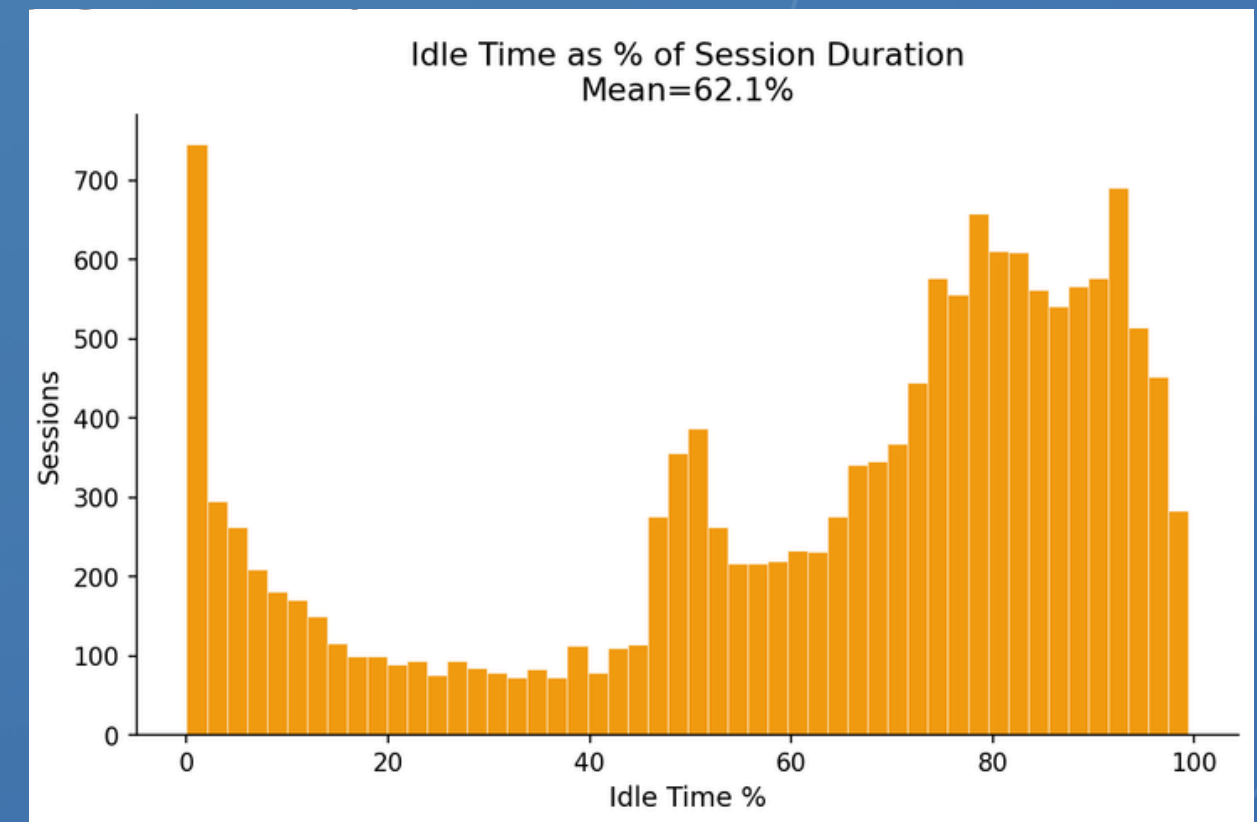
Revenue leakage — off-peak capacity wasted at ₹15/kWh regardless of demand



62.1% of session time is idle (car parked after charging done) — charger blocked for nothing

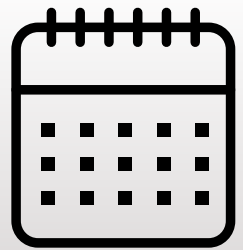


UrbanEV: mean utilization is only ~27–34% — massive underuse that discounts could unlock



DATA & EDA — ACN CHARGING PATTERNS

15K sessions, Caltech/JPL campus, Apr–Dec 2018



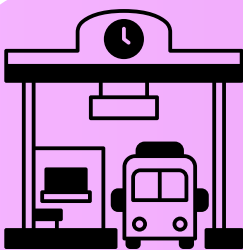
Mon–Fri = ~85% of sessions — workplace charging dominates; weekends are quiet



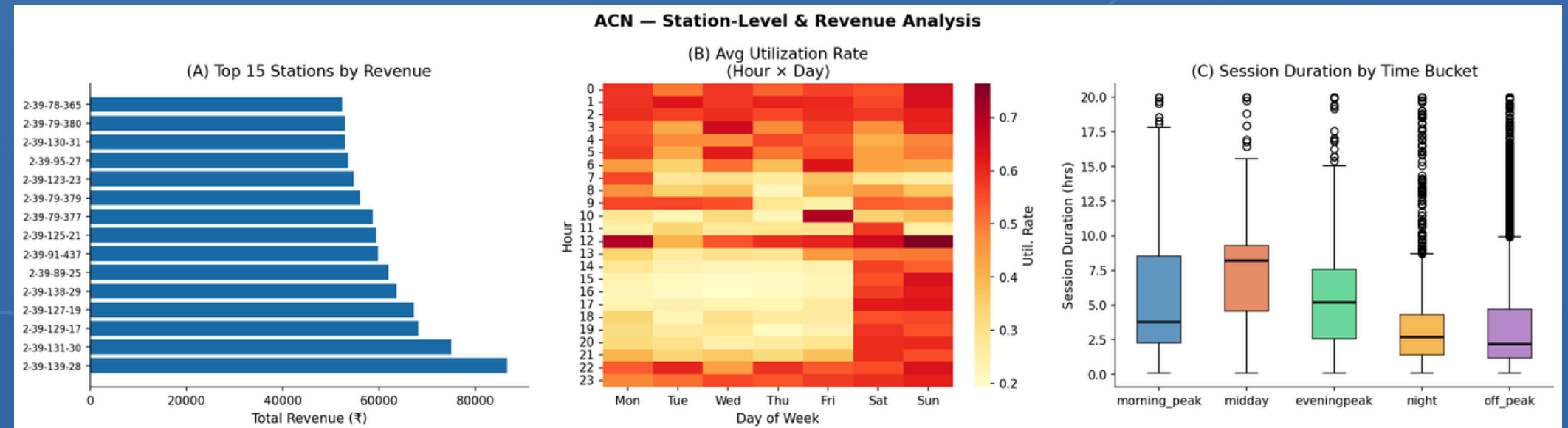
Evening peak (15–17h) is the busiest hour; overnight sessions are longest



Median session = 7.4 kWh — users charge partially, not to full battery



Top station (2-39-139-28) earns ~₹87K — revenue is highly concentrated in a few stations



DATA & EDA — URBAN EV SHENZHEN GRID

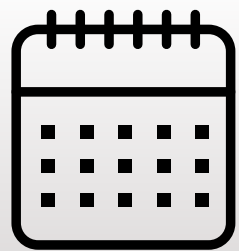
247 zones × 8,640 timestamps,
Jun–Jul 2022



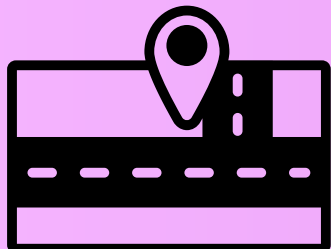
Max mean utilization ~34% —
network is far below the 80%
surge threshold at all hours



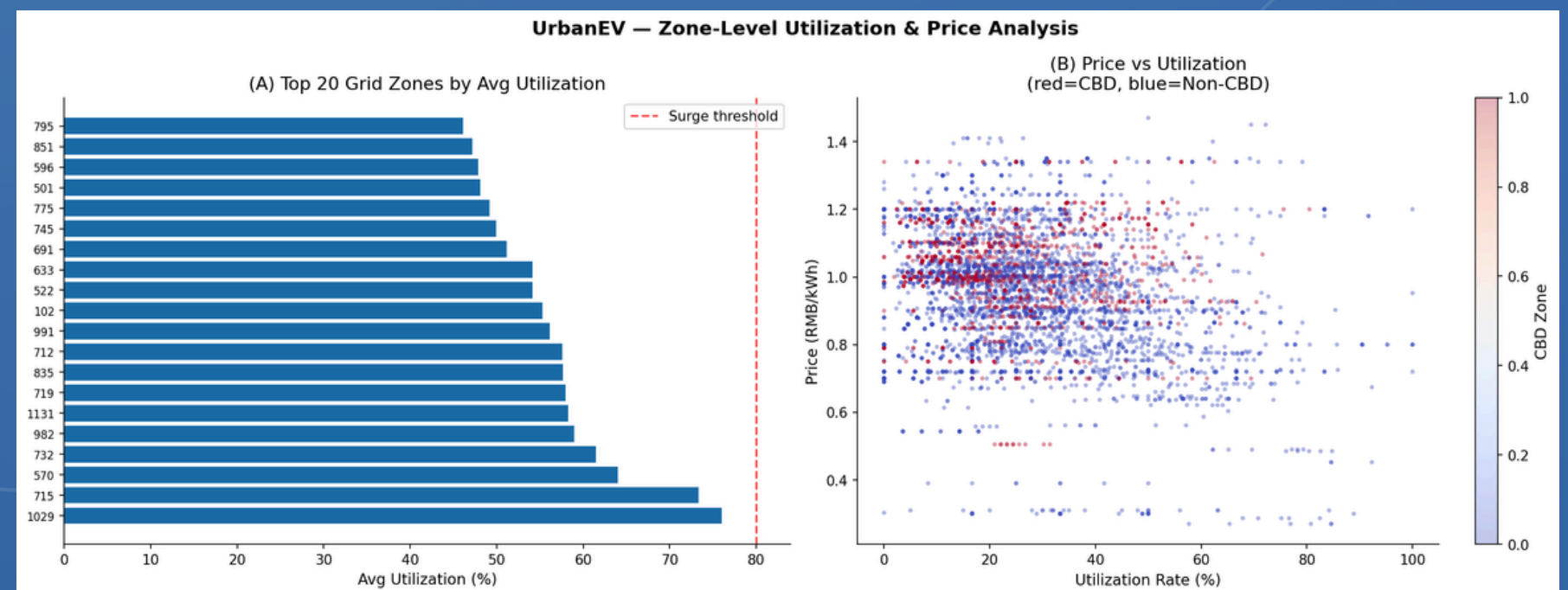
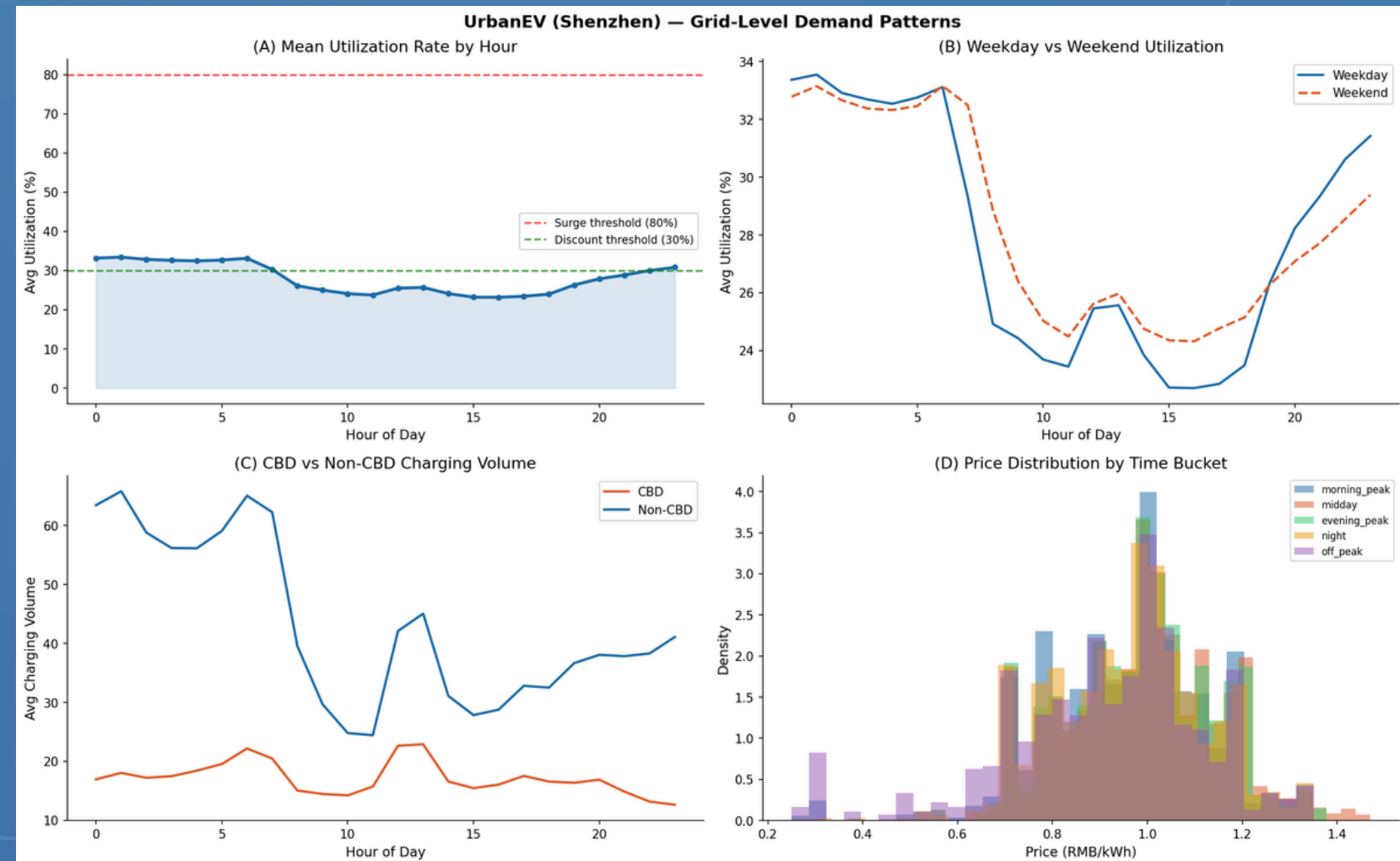
CBD mean 24.3% vs Non-CBD
29.3% — suburban zones are
actually busier than city centre



Weekday utilization slightly higher;
weekend drops uniformly across
all hours



Zone 1029 hits ~78% — closest
to surge trigger; only a handful of
zones approach peak threshold



AGENT 1 — DEMAND PREDICTION

Stacking ensemble: RF + XGBoost + LightGBM → Ridge meta-learner



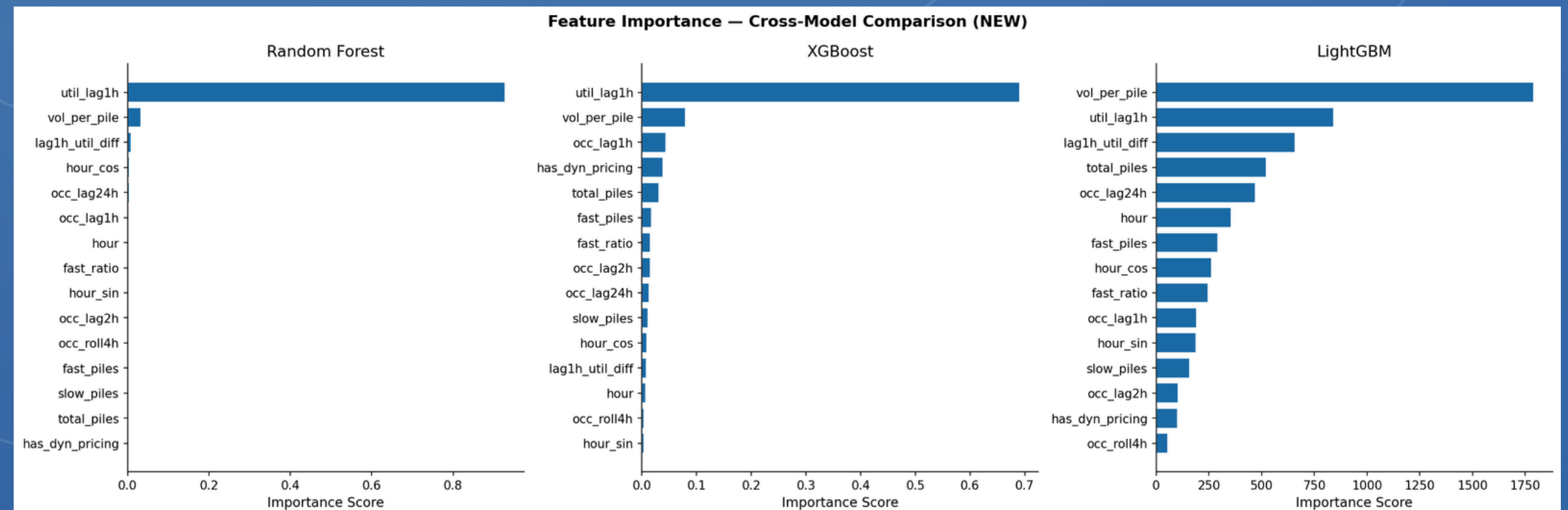
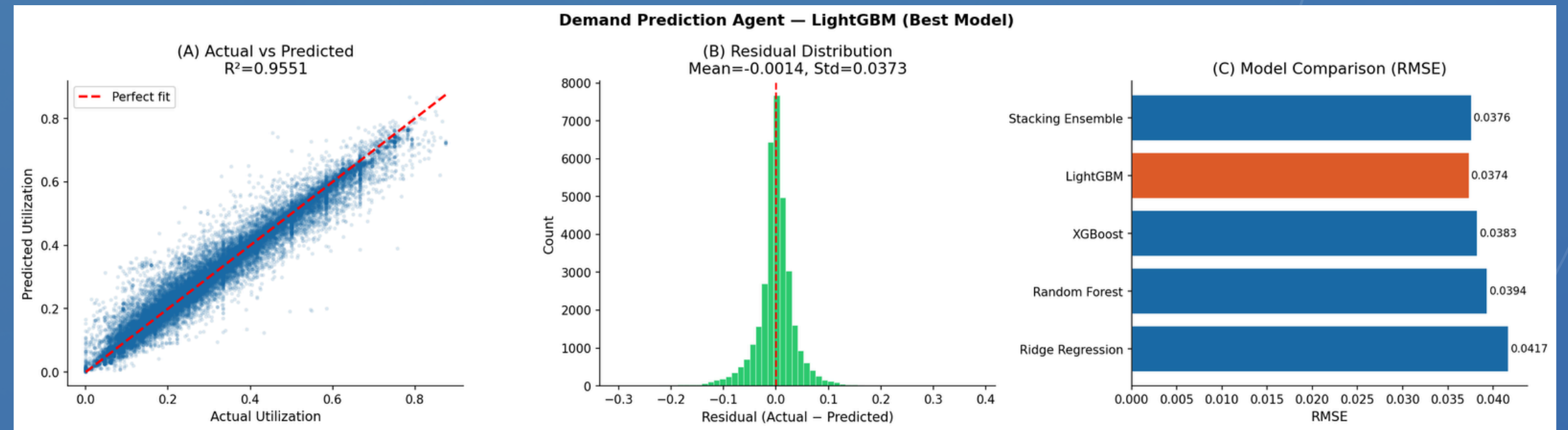
Best model: LightGBM —
RMSE 0.0374, $R^2 = 0.955$



- Residual mean = -0.0014 ,
 - Std = 0.037
- near-zero bias, tightly concentrated errors

Top features: util_lag1h (RF/XGB) and vol_per_pile (LightGBM) — recent history + congestion dominate

has_dyn_pricing appears in all 3 models — pricing itself is a demand signal



AGENT 2 — TARIFF PRICING AGENT

Elasticity-sweep grid search → dynamic ₹/kWh rules

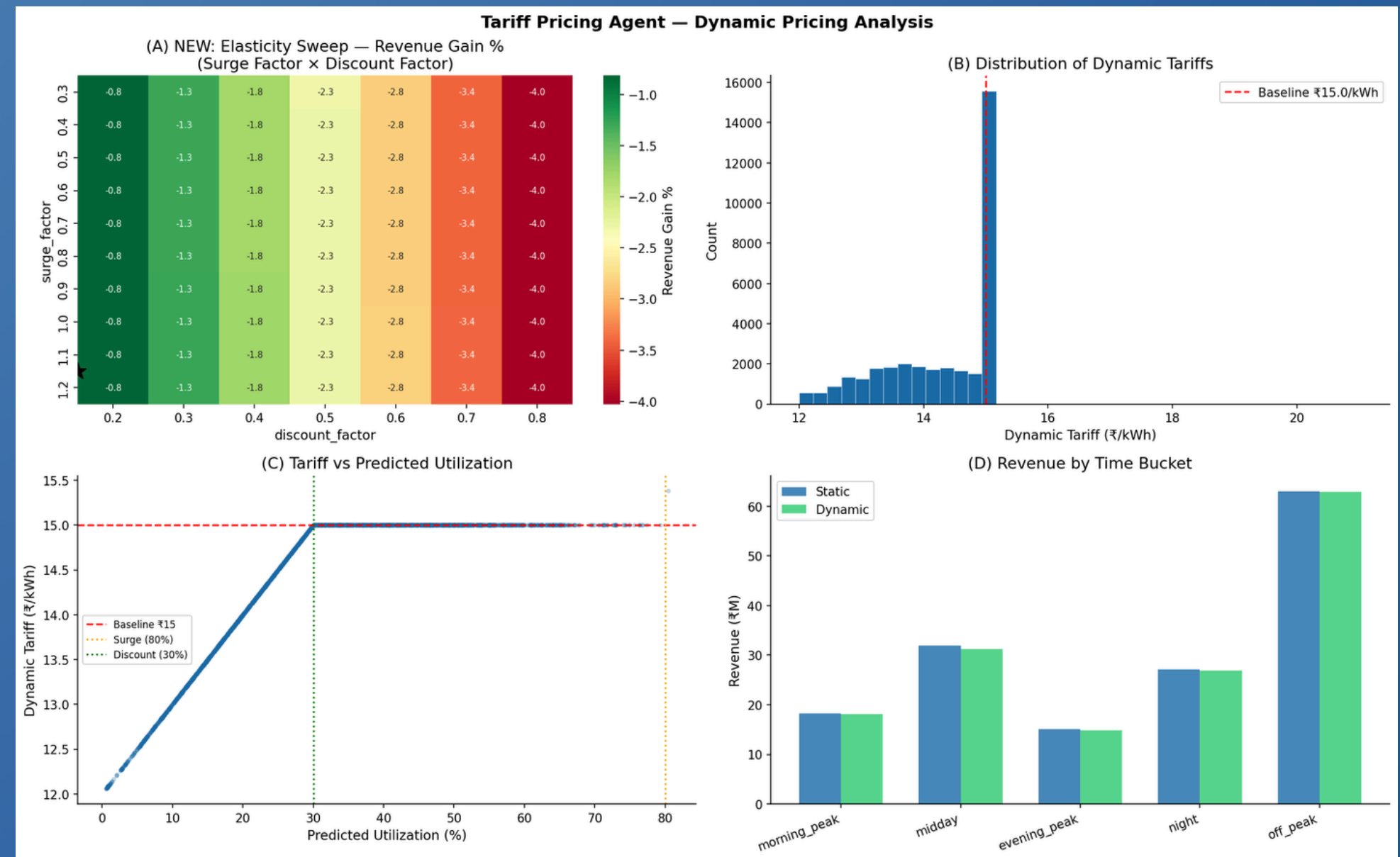
01. Surge zone (util > 80%): apply surge factor — converges to ~1.07× baseline

02. Discount zone (util < 30%): apply discount factor — calibrated to ~0.20

03. Night pricing gains the most: +18.45% revenue vs static

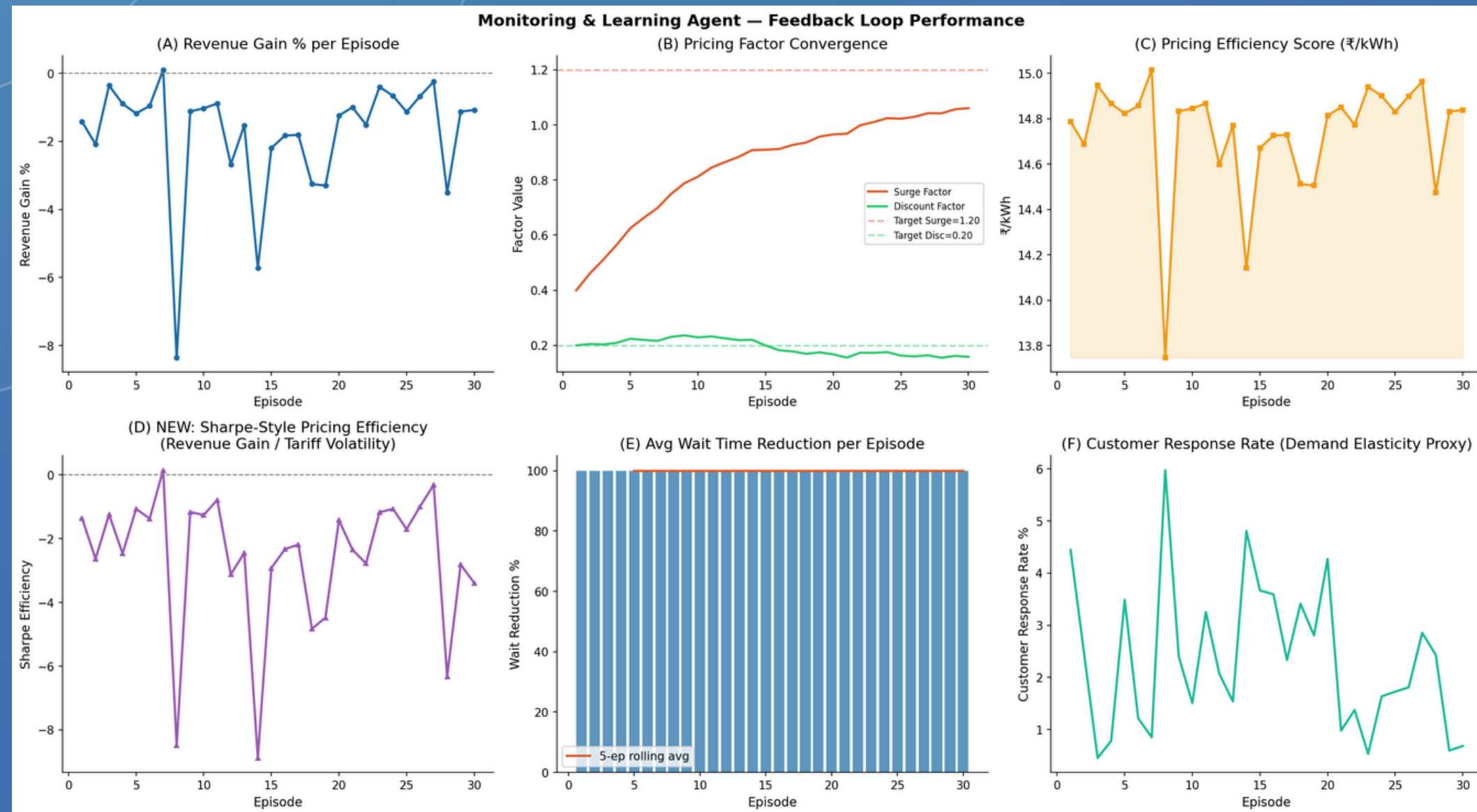
04. Morning peak +3.95%, Evening peak +6.84%, Midday -0.50%

05. Revenue gain most sensitive to discount_factor, not surge — UrbanEV rarely surges



AGENT 3 — MONITORING & LEARNING

30-episode feedback loop with self-correction



01. Surge factor converges from 0.4 → ~1.07 over 30 episodes (reaches target of 1.20)

02. Discount factor stable at ~0.20 — already at target from episode 1

03. Wait time reduction = 100% across all 30 episodes — charger availability maximised

04. Episode 8 outlier: revenue dip of -8.5% — system self-corrected within 2 episodes

05. Pricing efficiency score stabilises at ₹14.8–15.1/kWh band

Results, conclusions & future work

Results:

- Demand predicted with $R^2=0.9551$ — LightGBM outperforms stacking ensemble slightly
- Night-time dynamic pricing unlocks +18.45% revenue; overall system self-stabilises
- Feedback loop converges — 100% wait reduction, pricing efficiency stable at ~₹14.9/kWh

Future work:

- Add idle-time penalty fees to recover the 62.1% wasted charger time
- Incorporate real-time grid stress signals to trigger surge more aggressively
- Multi-agent coordination across zones to prevent demand displacement

0.955

R^2 model

100%

Wait cut

18.4%

Night gain

30

Episodes

0.037

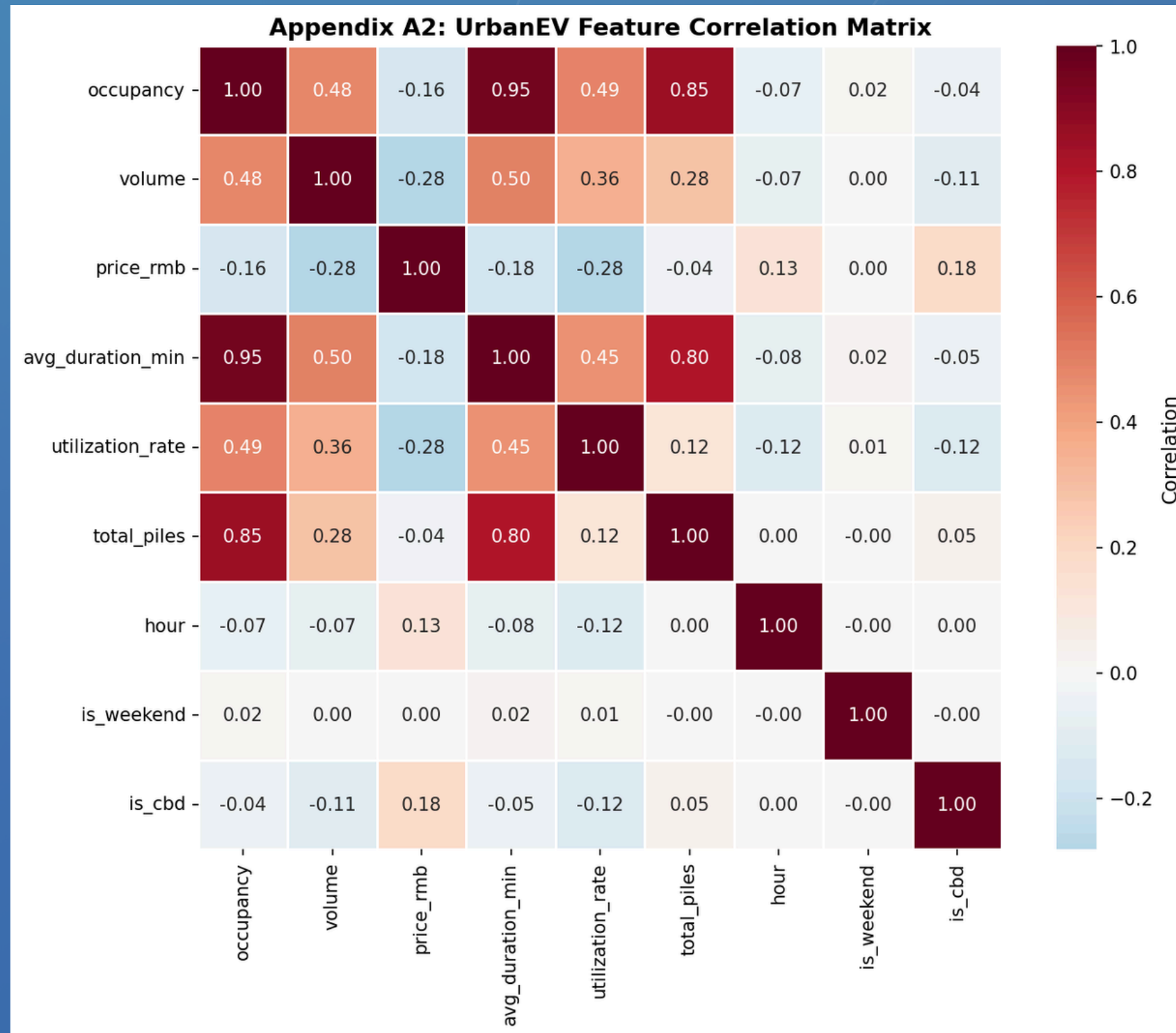
Best RMSE

62.1%

Idle — opp.

“Dynamic tariffs don't just earn more — they make the grid smarter.”

Appendix slides:

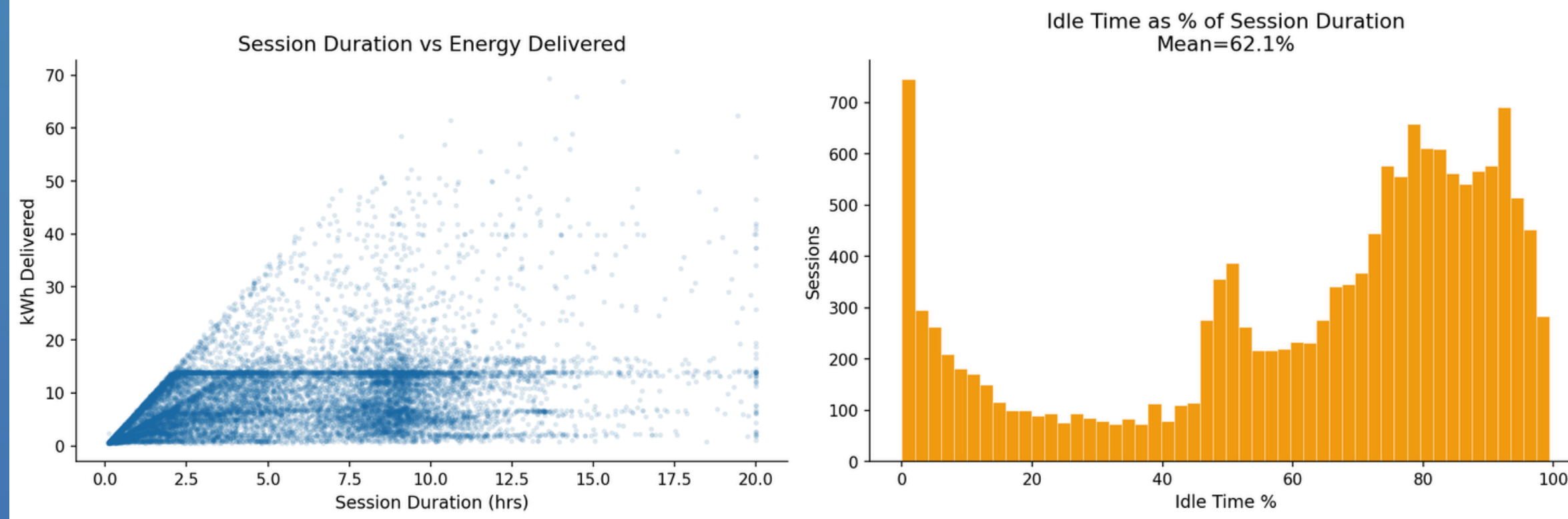


Key call-out:

- occupancy ↔ avg_duration = 0.95
- price ↔ volume = -0.28 (elasticity exists)

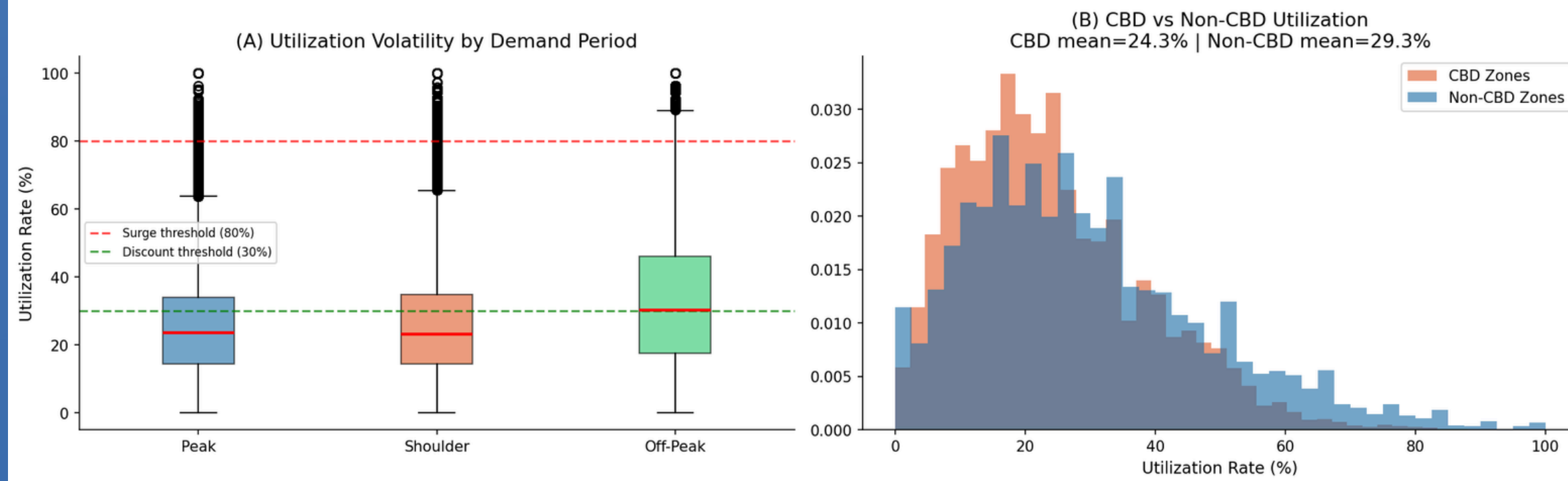
Appendix slides:

Appendix A3: ACN Charging Behavior Deep Dive



Full 2-panel chart — session duration vs kWh scatter + full idle time distribution

Demand Volatility — Peak vs Off-Peak vs Shoulder



Off-peak has widest spread —
unpredictable demand needs
adaptive pricing most

Thank You